

Why the F_{30} Periodic Brake Candidate Nearly Reaches the Rational 3:4:5 Burrau Initial Condition

September 5, 2026

Evidence status. The orbit discussed here is an ordinary high-precision numerical shooting solution, supported by independent integrations and a global Levi–Civita coordinate calculation. A validated interval proof that the exact periodic root exists has not yet been completed. It is therefore a very strong numerical candidate, not a theorem or a Pythagorean counterexample.

1. The exact rational target

Normalize the longest side and its opposite mass to one. The 3:4:5 Burrau initial condition is

$$(m_1, m_2, m_3) = \left(\frac{3}{5}, \frac{4}{5}, 1\right) = (0.6, 0.8, 1),$$

$$q_1 = \left(-\frac{1}{2}, 0\right), \quad q_2 = \left(\frac{1}{2}, 0\right), \quad q_3 = \left(\frac{7}{50}, \frac{12}{25}\right) = (0.14, 0.48),$$

with all three velocities zero. Every number in these normalized launch data is rational. The opposite side lengths are exactly $3/5$, $4/5$, and 1, and

$$\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 = 1.$$

In the Euclid parameterization this point is $u = 1/2$, or $u = 1/3$ after exchanging the legs.

2. Where F_{30} came from

Li and Liao searched for free-fall periodic brake orbits while holding the masses fixed at the exact rational ratio $(0.6, 0.8, 1)$ and allowing the initial triangle to vary. One catalog orbit, F_{30} , began at

$$q_3 \simeq (0.1446319096, 0.4773197126),$$

only about 5.35×10^{-3} from the exact 3:4:5 launch vertex. Thus the first reason for the proximity is deliberate selection: the parent catalog slice already had the exact 3:4:5 masses, and F_{30} was the catalog member whose brake triangle lay nearest the mass–side-tied target.

We then varied the two independent mass ratios and solved five equations: three equations demanding a simultaneous second brake and two equations demanding that each varied mass equal its opposite initial side. The nearby shooting root is

$$(m_1, m_2, m_3) \simeq (0.594811646571005, 0.801774973317793, 1),$$

$$q_3 \simeq (0.144521106471, 0.476902140016), \quad \tau \simeq 6.28923382379.$$

Doubling the brake-to-brake segment gives the numerically periodic labelled orbit shown below.

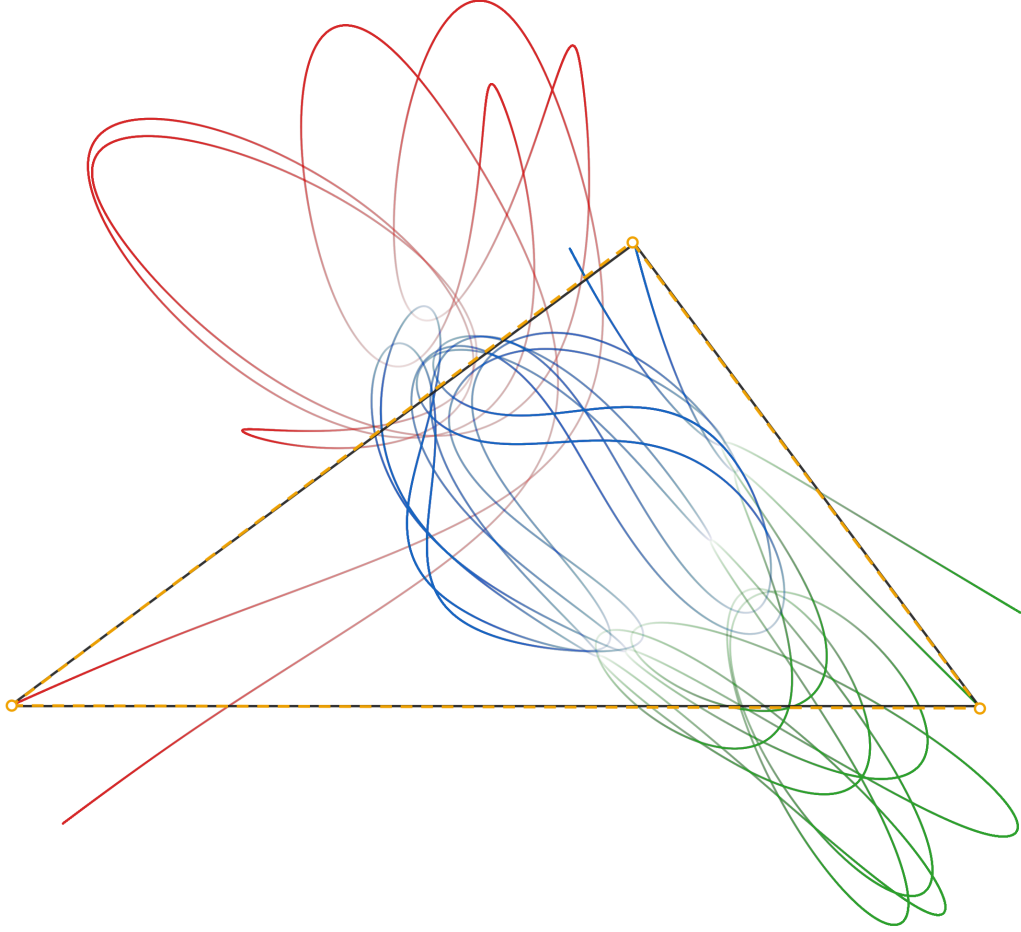


Figure 1: The F_{30} mass-side periodic candidate. Red, green, and blue trace the three labelled bodies at uniform increments of physical time. The solid dark triangle is the actual initial condition. The dashed gold triangle is the labelled 3:4:5 triangle after optimal translation, rotation, and one common scaling.

3. How close is it?

Quantity	Exact 3:4:5	F_{30} tied candidate	Difference
m_1	0.600000000000	0.594811646571	-0.005188353429
m_2	0.800000000000	0.801774973318	+0.001774973318
q_{3x}	0.140000000000	0.144521106471	+0.004521106471
q_{3y}	0.480000000000	0.476902140016	-0.003097859984
$m_1^2 + m_2^2 - 1$	0	-0.003355997265	-0.003355997265
largest initial angle	90°	90.2015966°	+0.2015966°

The two leg masses differ from 3/5 and 4/5 by approximately -0.865% and +0.222%. The candidate's leg-vector length is

$$\sqrt{m_1^2 + m_2^2} \simeq 0.9983205912,$$

so the unit side is only about 0.168% longer than the hypotenuse that would make these particular two legs exactly right.

For the figure, let z_i be the complex vertices of the exact labelled 3:4:5 triangle and w_i the actual

launch vertices. The displayed reference minimizes

$$\sum_{i=1}^3 |w_i - (b + \alpha(z_i - \bar{z}))|^2$$

over the translation b and the complex multiplier α , which supplies one rotation and one positive common scale but no reflection or relabeling. The optimal scale is

$$|\alpha| \simeq 0.9991493750,$$

and the RMS vertex mismatch after alignment is only

$$2.23745 \times 10^{-3}.$$

4. Why a small continuation was enough

Away from collision, a nondegenerate periodic shooting root varies smoothly when the masses vary. The original F_{30} triangle already nearly satisfied the two mass-side equations. Allowing both independent mass ratios to move therefore supplied exactly two controls with which to drive those two side residuals to zero while the three second-brake equations were maintained.

The orbit also has a long sequence of syzygies and an extremely close binary passage, with a resolved separation of order 5×10^{-4} . Such passages make the return map highly sensitive: small changes in the launch geometry or masses can produce large endpoint changes. This helps explain why modest parameter adjustments can repair several shooting residuals. It is a dynamical interpretation, however, not a theorem that forces an intersection with the right-triangle locus.

The resulting mass-side solution appears locally isolated and transverse. After the three brake equations and two mass-side equations have been imposed, the right-angle equation

$$m_1^2 + m_2^2 = 1$$

is one additional condition with no remaining continuous tuning parameter. No known symmetry forces it. This is why the orbit gets remarkably close but then stops at a defect of approximately -0.003356 .

5. Is the F_{30} initial condition rational?

No rationality result is known for the continued F_{30} root. Its printed masses and coordinates are numerical approximations to a shooting root. Finite decimal approximations can prove neither rationality nor irrationality.

There are three different statements that should not be conflated:

1. The *parent mass slice* $(3/5, 4/5, 1)$ is exactly rational.
2. The original catalog orbit on that slice has numerical launch coordinates; those coordinates are not known to be rational.
3. The continued mass-side candidate has numerical masses $(0.594811\dots, 0.801775\dots, 1)$ and numerical coordinates. They are not known to be rational or irrational.

Moreover, the continued candidate is certainly not an exact Pythagorean initial condition at the level of the computed root: its triangle is obtuse, not right. Even proving that one or both of its masses were rational would not repair the nonzero Pythagorean defect.

One may project the candidate's mass pair radially onto the unit Pythagorean circle. Its angular direction then corresponds to

$$u_{\text{proj}} = \frac{m_2}{\sqrt{m_1^2 + m_2^2} + m_1} \simeq 0.5032695682,$$

near the rational value $1/2$. But this is only a quantitative description of proximity: because the unprojected candidate is not on the unit circle, it has no actual Euclid parameter u .

Conclusion

F_{30} is close to a rational Pythagorean initial condition for two concrete reasons: it was selected from the exact rational 3:4:5 mass slice, and a smooth two-mass continuation then imposed the two mass-side constraints with only a small displacement. Its complicated close-encounter dynamics gives the shooting map enough sensitivity to make this adjustment possible. What is still absent is an exact mechanism forcing the sixth condition—the right angle—or forcing the resulting root to have rational arithmetic data.